

IsaacLab-Arena on NVIDIA Brev

Workshop organizer guide: build the Launchable, test it, and share it with participants

Tanna TechBiz Physical AI Lab · September 2026 · Arena release/0.2.1, Isaac Sim 6.0.0, Isaac Lab 3.0.0

1. What you are building

A Brev **Launchable** is a one-click template: participants open a link, pick a GPU, and get a fully prepared machine. Ours gives every participant:

- A GPU node with the IsaacLab-Arena repository (release/0.2.1, submodules included) already cloned under `~/IsaacLab-Arena`.
- The Arena Docker image already built, so `./docker/run_docker.sh` starts in seconds instead of an hour.
- The standalone Isaac GR00T checkout and its uv environment (steps 9-12 of the NVIDIA workflow page).
- A GPU-accelerated Linux desktop viewable **in the browser** (noVNC on port 6080), so the Isaac Sim window is visible without any local install.
- Optionally JupyterLab with the participant notebook.

All of this is done by one file, `brev-setup.sh`, which you paste into the Launchable as its setup script. The participant-facing steps follow NVIDIA's *GR00T End-to-End Workflow* → *Simulation Workflow* → *Setup: Isaac Lab-Arena* page.

Component	Where it runs	How participants reach it
XFCE desktop on Xorg :0 (NVIDIA virtual display)	Host VM	Browser, via the Secure Link for port 6080
x11vnc (password protected, localhost only)	Host VM, port 5900	Not exposed; noVNC proxies to it
noVNC / websockify	Host VM, port 6080	Secure Link (Brev login) or direct IP
Arena container <code>isaac_lab_arena:latest</code>	Docker, <code>--net=host</code> , <code>DISPLAY=:0</code>	Terminal on the desktop
Isaac-GR00T (uv env)	Host VM, <code>~/Isaac-GR00T</code>	Terminal on the desktop
JupyterLab (optional)	Host VM, port 8888	Brev's built-in Jupyter Secure Link

2. Before you start

- **Brev organization** with billing enabled and enough credits. Budget roughly **USD 2.5-3.5 per**

participant-hour for an RTX PRO 6000 / L40S class GPU, plus the build time.

- **GPU choice.** Isaac Sim needs an RTX-capable GPU with 24 GB or more. Good choices: RTX PRO 6000, L40S, A10G (48 GB variants). Set **disk to 300 GiB or more**: the Arena image plus Isaac Sim caches take 60-80 GB.
- **Files from this kit:** `brev-setup.sh`, `IsaacLab-Arena-Workshop-Participant-Guide.pdf`, `isaacsim_arena_workshop.ipynb`.
- **Time.** A fresh deploy of this Launchable takes **60-90 minutes** to reach "Built" because it builds the Arena image. Deploy the day before, or at least 2 hours before the session.

3. Step 1 - Test the setup script on a dev instance (recommended)

Run the script once on any running Brev VM-mode instance before baking it into the Launchable. This catches driver or registry problems early.

```
brev copy brev-setup.sh <instance-name>:~/brev-setup.sh
brev shell <instance-name>
# on the instance:
VNC_PASSWORD='choose-a-password' sudo -E bash ~/brev-setup.sh
# or in the background with a log:
VNC_PASSWORD='choose-a-password' nohup sudo -E bash ~/brev-setup.sh > ~/setup.out 2>&1 &
tail -f /var/log/arena-workshop-setup.log
```

When it prints `=== DONE`, read `~/WORKSHOP.md` (it contains the URL and password), then confirm:

```
ss -tlnp | grep 6080 # noVNC listening
DISPLAY=:0 glxinfo | grep "OpenGL renderer" # must name the NVIDIA GPU, not llvmpipe
docker images | grep isaacsim_arena # image built
cd ~/IsaacLab-Arena && git submodule status # no leading "-" on any line
```

Registered / BYOC nodes. Instances registered from your own cloud (Massed Compute, Shadeform, etc.) do not show Secure Link or tunnel controls in the Access tab. On those, run the script with `NOVNC_TLS=1`: it serves noVNC over self-signed HTTPS on the node's public IP and opens port 6080 in UFW (and in the node's `init.sh`, which resets UFW at boot). Participants accept the certificate warning once and enter the VNC password.

4. Step 2 - Create the Launchable

In the Brev console go to **Launchables** → **Create Launchable** and fill in the sections in order.

4.1 Details

- **Name:** `isaacsim-arena-workshop` (cannot be changed later). **Description:** "IsaacLab-Arena 0.2.1 + Isaac Sim 6.0 with browser desktop, for the GR00T simulation workflow workshop".

4.2 Default Hardware Configuration

- GPU: RTX PRO 6000 or L40S, 1 GPU. Participants may pick another RTX-class card at deploy time.
- Disk: **300 GiB**.

4.3 Software Configuration

- Mode: **VM Mode** (Python, CUDA and Docker are preinstalled; the script relies on Docker and the NVIDIA driver).
- Toggle **Jupyter** on if you want the notebook available at port 8888.
- **Setup script**: paste the full contents of `brev-setup.sh`.

4.4 Source

- Choose **Notebook** and upload `isaac lab_arena_workshop.ipynb`, or choose **None** if you will distribute the notebook another way.

4.5 Network

Setting	Value	Why
Secure Link	Name <code>desktop</code> , port 6080	Browser desktop behind Brev login. The name becomes part of the URL.
Secure Link (optional)	Name <code>jupyter</code> , port 8888	Only if Jupyter was not enabled in 4.3 (that toggle creates its own link).
TCP/UDP ports	none	Isaac Sim's WebRTC stream is not used; VNC needs nothing extra.

4.6 Launch Parameters

These become environment variables for the setup script. Participants see them on the deploy page.

Parameter	Type / required / default	Purpose
<code>VNC_PASSWORD</code>	Text, required, no default	Password typed into noVNC. Use one shared workshop password, or let each participant set their own.
<code>PREBUILD_IMAGE</code>	Choice 1 / 0, optional, default 1	Build the Arena image during setup (recommended).
<code>INSTALL_GROOT</code>	Choice 0 / 1, optional, default 0	Also build the GROOT-flavored image (adds about 45 min). Not needed for the sim-setup module.
<code>NOVNC_TLS</code>	Choice 0 / 1, optional, default 0	Keep 0 behind a Secure Link. Set 1 only for

BYOC nodes with no Secure Link.

<code>ARENA_BRANCH</code>	Text, optional, default release/0.2.1	Pin the Arena version used in the workshop.
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4.7 View Access

- **Anyone with the link** for an external workshop, or **Only my organization** if all participants are org members.

Click **Create Launchable**, then copy the share URL from the Launchable page.

5. Step 3 - Dry run from the share link

1. Open the share URL in a private browser window, exactly as a participant would. Click **Deploy Launchable**, fill in `VNC_PASSWORD`, confirm the GPU.
2. Watch the instance page. Status goes **Deploying** → **Building** → **Built**. Building is where the setup script runs; expect 60-90 minutes on the first deploy.
3. Open the **Logs** tab if Building takes longer than 2 hours. The script also writes `/var/log/arena-workshop-setup.log`.
4. When Built, click the **desktop** Secure Link in the Access tab, log in with your Brev account, enter the VNC password. You should see an XFCE desktop.
5. Open a terminal on the desktop and run the smoke test from the participant guide, section 4 (policy_runner with `--viz kit`). An Isaac Lab window must appear on the desktop within 3-5 minutes on the first launch.
6. Stop or delete the test instance afterwards to save credits.

If the dry run passes, the Launchable is ready. Nothing else needs to be rebuilt; every deploy repeats the same script.

6. Step 4 - Share with participants

- **Send** the share URL, the participant PDF, and the notebook. Tell participants the VNC password policy (shared password or self-chosen).
- **Accounts.** Each participant needs a Brev account to log in through the Secure Link. If they deploy their own instances they also need to be in an org with credits: invite them with `brev invite` or the console's org settings.
- **Who pays.** Option A: participants deploy the Launchable into their own org (they pay). Option B: you deploy N instances into your org ahead of time and hand each participant an instance name plus its desktop URL; use **Share SSH Access** → **Add User** on each instance if they need a shell outside the desktop. Option B avoids the 60-90 minute wait on the day.
- **Pre-deploy timing.** Start instances at least 2 hours before the session. Once Built, they stay ready as long as they are running; stopping and starting keeps the built image and repo.
- **Reboots.** The desktop is a systemd service (`gpu-desktop`) and comes back automatically. Brev does not rerun the setup script on restart, and it does not need to.

7. Day-of runbook

When	Action
T-2h	All instances Running and Built. Open each desktop link once to confirm the XFCE screen appears.
T-30m	On each instance, start the container once so caches are warm: <code>cd ~/IsaacLab-Arena && ./docker/run_docker.sh</code> , then <code>exit</code> .
T-0	Participants open the desktop link, enter the VNC password, open a terminal, follow the participant guide from section 2.
During	Keep this guide's troubleshooting table handy. <code>tail -f /var/log/arena-workshop-setup.log</code> and <code>/tmp/xorg.log</code> are the two logs that matter.
After	Stop instances (keeps disk, stops GPU billing) or delete them.

8. Troubleshooting

Symptom	Cause	Fix
Secure Link shows "connection refused" or a blank page	noVNC not running	On the instance: <code>sudo systemctl restart gpu-desktop</code> ; check <code>/tmp/xorg.log</code> for "(EE)".
Xorg log: "no screens found" / "UseDisplayDevice None is not supported"	Server-class GPU needs the driver's virtual-display mode	The generated <code>xorg.conf</code> already omits <code>UseDisplayDevice</code> . If you edited it, keep only <code>AllowEmptyInitialConfiguration</code> and the <code>Virtual</code> line.
<code>glxinfo</code> shows <code>llvmpipe</code>	NVIDIA X driver package missing	<code>sudo apt install xserver-xorg-video-nvidia-<driver major></code> , then restart the service.
noVNC connects but shows a black screen	XFCE session died or screensaver	<code>~/start-desktop.sh</code> from an SSH shell; the script also disables DPMS.
Build phase fails at <code>git submodule</code>	SSH URLs in <code>.gitmodules</code>	The script sets a global URL rewrite to HTTPS. Re-run <code>git submodule update --init --recursive</code> as the login user.
Docker build fails pulling <code>nvr.io/nvidia/isaac-sim:6.0.0-dev2</code>	Registry hiccup; the image is public	Re-run: <code>cd ~/IsaacLab-Arena && ./docker/run_docker.sh -r</code> .
<code>policy_runner</code> exits with "required: --policy_type" or	Argument order / missing budget	Runner options (<code>--policy_type zero_action --viz kit --num_steps 3000</code>) go before the environment

"num_steps or num_episodes must be provided"		name; environment options after it.
Isaac Sim window opens but is very slow through the browser	VNC frame rate	Use <code>resize=scale</code> in the noVNC URL and set quality to 6 in the noVNC side panel. Advise 1920x1080 or lower browser windows.
Apple falls through the shelf in the test	Known first-run asset warm-up	Re-run the test once (documented by NVIDIA).
Participant cannot log in to the Secure Link	No Brev account or wrong org	They must sign in with the same account that was invited; otherwise use the direct-IP mode (<code>NOVNC_TLS=1</code>).

9. Security notes

- VNC is password protected and bound to localhost; only noVNC on 6080 is reachable, and behind a Secure Link it also requires Brev login.
- The Arena container runs `--privileged --net=host` by design of `run_docker.sh`; treat every participant instance as single-tenant.
- Change the shared VNC password after the workshop or delete the instances.

10. Reference links

- NVIDIA workflow page participants follow: <https://docs.nvidia.com/learning/physical-ai/gr00t-e2e-workflow/latest/simulation-workflow/sim-setup-isaac-lab-arena.html>
- Brev Launchables concepts: <https://docs.nvidia.com/brev/concepts/launchables>
- Brev console reference: <https://docs.nvidia.com/brev/guides/console-reference>
- IsaacLab-Arena repository: <https://github.com/isaac-sim/IsaacLab-Arena> (tag release/0.2.1)
- IsaacLab-Arena docs: <https://isaac-sim.github.io/IsaacLab-Arena/main/>